

1. Introduction

Currency Trader Pro operates exclusively in **Paper Trading Mode**. All trades use a virtual \$10,000 starting balance. No real exchange accounts are connected and no real money is ever at risk.

Capability	Detail
Live Crypto Prices	Binance WebSocket — BTC, ETH, SOL, BNB, ADA, DOGE, XRP, AVAX
Forex Prices	Simulated realistic prices — EUR/USD, GBP/USD, USD/JPY + 4 more
Order Types	Market, Limit, Stop — with optional Take-Profit & Stop-Loss
Auto Trading	4 built-in strategies run every 30 seconds automatically
Portfolio Analytics	Real-time P&L, allocation chart, 30-day daily P&L history
Trade History	Full audit trail with per-trade fee and realised P&L
Executable	Single binary — no installation, no database server required

1.2 System Requirements

User Guide & Reference Manual

Component	Minimum	Recommended
Operating System	Windows 10 / Debian 11	Windows 11 / Debian 12
CPU	Dual-core 2 GHz	Quad-core 3 GHz
RAM	2 GB	4 GB
Disk Space	150 MB	500 MB
Network	Required (market data)	Broadband
Browser	Chrome / Firefox / Edge	Chrome 120+

Note: An internet connection is required to receive live cryptocurrency prices from Binance. Forex prices are generated locally and work offline.

2. Installation & First Launch

2.1 Using the Pre-Built Executable

The simplest way to run Currency Trader Pro is using the pre-built binary. No Python or Node.js installation is required.

Linux / Debian

```
# 1. Download or copy currency_trader to your machine # 2. Make it executable chmod +x currency_trader # 3. Run it ./currency_trader
```

The application will start a local web server on port **8765** and automatically open your default browser. If the browser does not open, navigate manually to: **http://localhost:8765**

Windows

```
# Double-click currency_trader.exe # OR run from Command Prompt: currency_trader.exe
```

Windows Defender SmartScreen may show a warning on first run because the binary is not code-signed. Click **More info** → **Run anyway** to proceed.

2.2 Building from Source

If you have Python 3.10+ and Node.js 18+ installed, you can build the executable yourself from the source code.

Linux / macOS

```
cd currency_trader chmod +x build.sh ./build.sh # Output: dist/currency_trader
```

Windows

```
cd currency_trader build.bat # Output: dist\currency_trader.exe
```

The build script automatically:

- Creates a Python virtual environment and installs all backend dependencies
- Installs Node.js packages and compiles the React frontend
- Bundles everything into a single self-contained executable via PyInstaller

2.3 Development Mode

For development with hot reload on both frontend and backend:

```
cd currency_trader ./run_dev.sh # Backend → http://localhost:8765 (FastAPI) # Frontend → http://localhost:5173 (Vite dev server)
```

■ **Warning:** Development mode requires Python 3.10+ with virtual environment support and Node.js 18+. The `run_dev.sh` script installs all dependencies automatically.

3. Interface Overview

The interface is divided into four main zones:

Zone	Description
Ticker Bar (top)	Live scrolling price feed for all 15 symbols with 24h change
Sidebar (left)	Navigation menu, mini portfolio summary, and connection status
Main Content Area	The active page — changes based on the sidebar selection
Status Bar (bottom)	Page number and paper trading mode indicator

3.1 Navigation Pages

Page	Icon	Purpose
Dashboard	■	Overview — chart, portfolio stats, top movers, P&L history
Trade	■	Place buy/sell orders, view and cancel open orders
Portfolio	■	Positions, asset allocation, performance statistics
Auto Trading	■	Create and manage automated algorithmic strategies
History	■	Full trade history, open orders, win rate
Settings	■	App configuration, data sources, portfolio reset

3.2 Live Ticker

The scrolling banner at the top of every page shows all 15 supported symbols. Each entry displays the current price and 24-hour percentage change. Green values indicate a price increase; red indicates a decrease. Hover over the ticker to pause scrolling.

3.3 Portfolio Summary (Sidebar)

The sidebar always shows your current total portfolio value and overall P&L; so you can monitor performance at a glance without leaving the current page.

The **Live Feed** indicator at the bottom of the sidebar shows the WebSocket connection status. When connected, prices update every second in real time.

4. Dashboard

The Dashboard is the default landing page and provides a high-level view of your account and the market.

4.1 Statistics Cards

Card	Description
Portfolio Value	Total account value (cash + open position market value)
Cash Balance	USD available for new trades
Holdings Value	Current market value of all open positions
Selected Symbol	Live price and 24h change for the active chart symbol

4.2 Candlestick Chart

The main chart area displays a **1-minute OHLCV candlestick chart** powered by TradingView's Lightweight Charts library. Features include:

- Pan and zoom with mouse scroll or trackpad pinch
- Crosshair with exact OHLC values on hover
- Volume histogram in the lower portion of the chart
- Green candles = bullish (close $\geq$ open) | Red candles = bearish (close $<$ open)
- Live tick updates — the latest bar updates in real time as new prices arrive

Use the **symbol dropdown** above the chart to switch between any of the 15 available symbols. The chart automatically reloads with up to 200 historical bars.

4.3 Top Movers

The right panel lists the 6 symbols with the largest absolute 24-hour price movement. Click any symbol to instantly switch the chart to that symbol.

4.4 Daily P&L; Sparkline

An area chart shows your realised P&L; aggregated by calendar day for the past 14 days. This helps identify performance trends and trading patterns over time.

4.5 Open Positions

If you have any open positions they appear in a compact list in the sidebar area of the dashboard showing symbol, quantity, current value, and unrealised P&L.;

5. Trading

The Trade page allows you to place orders manually. The chart and order form are displayed side-by-side for efficient decision-making.

5.1 Order Types

Type	Behaviour
Market	Executes immediately at the current live price. No price field required.
Limit	Placed in the order book and fills only when the market price reaches your specified limit price. Buy fills when price
Stop	Triggers when the market reaches the stop price, then executes at market. Useful for stop-loss protection on existi

5.2 Placing an Order

Follow these steps to place a trade:

Step	Action
1	Select a symbol from the dropdown above the chart, or navigate to the Dashboard and click a symbol in Top Movers.
2	Choose Buy or Sell using the tab buttons (green/red).
3	Select the Order Type (Market, Limit, or Stop).
4	If using Limit or Stop, enter the trigger price in the price field.
5	Enter the quantity. Use the slider + calc button to auto-fill a percentage of your available USD balance.
6	Optionally set Take-Profit and/or Stop-Loss prices.
7	Click the Buy/Sell button. A confirmation message appears below the form.

5.3 Quantity Calculator

The slider above the Quantity field lets you specify what percentage of your available USD balance to use for the trade. After adjusting the slider, click **calc** to fill the Quantity field automatically based on the current price.

Note: The estimated trade value is shown in small grey text below the quantity field as you type, giving you an immediate cost preview before submitting.

5.4 Take-Profit & Stop-Loss

When provided, Take-Profit and Stop-Loss prices are stored with your position. The trading engine monitors these levels continuously:

- **Take-Profit:** When the market price reaches or exceeds this level, the system automatically places a market SELL order to lock in profit.
- **Stop-Loss:** When the market price falls to or below this level, the system places a market SELL order to limit your loss.

Note: TP/SL are stored as reference prices and checked every 0.5 seconds against the live price. For SELL orders, the logic is reversed (SL above entry).

5.5 Managing Open Orders

Open (unfilled) Limit and Stop orders appear in a list below the order form. Click the X button next to any order to cancel it immediately. Market orders are always filled instantly and never appear in the open orders list.

5.6 Fees

A flat **0.1% taker fee** is applied to every executed order. Fees are deducted from your USD balance on buys and from the proceeds on sells. The fee is shown separately in Trade History so you can track total costs.

6. Portfolio

The Portfolio page provides a detailed breakdown of your account, open positions, and performance metrics.

6.1 Summary Cards

Card	Formula
Total Value	Cash Balance + Holdings Value (at current market prices)
Cash Balance	USD available — decremented on buys, incremented on sells
Total P&L	Total Value – Initial Balance (\$10,000)
Return %	$(\text{Total Value} / 10,000 - 1) \times 100$

6.2 Asset Allocation Chart

A pie chart shows the proportional allocation of your portfolio across cash and each open position (by current market value). This gives an instant view of concentration risk. Mouse over any segment to see the exact USD value.

6.3 Daily P&L; Bar Chart

A bar chart of realised P&L; grouped by calendar day over the last 30 days. Green bars indicate profitable days; red bars indicate losing days. Only closed (SELL) trades contribute to realised P&L.;

6.4 Performance Statistics

Metric	Description
Total Trades	Count of all executed trades (both buys and sells)
Win Rate	Percentage of SELL trades with positive P&L
Avg Win	Average P&L of winning trades ($\text{P\&L} > 0$)
Avg Loss	Average P&L of losing trades ($\text{P\&L} < 0$), shown as positive
Total Fees	Sum of all fees paid across all trades

6.5 Open Positions Table

Column	Description
Symbol	Trading pair (e.g. BTC/USDT)
Qty	Number of units held
Avg Entry	Volume-weighted average purchase price
Current	Latest live market price

Column	Description
Value	$\text{Qty} \times \text{Current Price}$
Unrealised P&L	$(\text{Current} - \text{Avg Entry}) \times \text{Qty} - \text{fees accrued}$
%	Unrealised P&L as percentage of entry cost

Note: Unrealised P&L; updates every 5 seconds. The portfolio total refreshes automatically via WebSocket — no manual refresh needed.

7. Auto Trading

The Auto Trading page lets you configure algorithmic strategies that trade automatically on your behalf using your paper balance. Strategies run every **30 seconds** when active.

All automated strategies use **paper balance only**. They follow the same fee rules and portfolio limits as manual trades. No external exchange APIs are called.

7.1 Available Strategies

Strategy	Type	Buy Signal	Sell Signal
SMA Crossover	Trend-Following	Fast SMA crosses above Slow SMA	Fast SMA crosses below Slow SMA
RSI	Mean-Reversion	RSI falls below oversold threshold (default: 30)	RSI rises above overbought threshold (default: 70)
MACD	Momentum	MACD line crosses above Signal line	MACD line crosses below Signal line
Bollinger Bands	Mean-Reversion	Price touches or breaks below lower band	Price touches or exceeds upper band

7.2 Strategy Parameters

SMA Crossover

Parameter	Default	Description
fast_period	10	Number of bars for the fast (short-term) moving average
slow_period	30	Number of bars for the slow (long-term) moving average

RSI

Parameter	Default	Description
period	14	Number of bars used to calculate RSI
oversold	30	RSI level that triggers a BUY signal
overbought	70	RSI level that triggers a SELL signal

MACD

Parameter	Default	Description
fast	12	Fast EMA period
slow	26	Slow EMA period
signal	9	Signal line EMA period

Bollinger Bands

Parameter	Default	Description
period	20	Look-back period for the moving average and std deviation

Parameter	Default	Description
std_mult	2.0	Number of standard deviations for the band width

7.3 Risk Management Settings

Setting	Default	Description
Position Size %	10%	Percentage of current USD balance used per entry
Stop Loss %	2%	Auto stop-loss placed at $\text{entry_price} \times (1 - \text{SL}\%)$
Take Profit %	4%	Auto take-profit placed at $\text{entry_price} \times (1 + \text{TP}\%)$
Max Positions	1	Max concurrent open positions for this strategy

7.4 Creating a Strategy

Step	Action
1	Click New Strategy in the top-right corner of the Auto Trading page.
2	Enter a descriptive name (e.g. 'BTC RSI 14').
3	Choose a Strategy Type from the dropdown. A description appears below.
4	Select the Symbol to trade (e.g. BTC/USDT).
5	Adjust Position Size, Stop Loss, and Take Profit percentages.
6	Review and optionally edit the strategy-specific parameters.
7	Click Create . The strategy is saved but not yet active.
8	Click Start on the strategy card to activate it.

7.5 Strategy Logs

Each strategy card has a **Logs** button that expands a scrollable log panel showing the last 20 actions. Log entries include:

- Buy/Sell decisions with price, quantity, and the signal that triggered them
- Errors (shown in red) if an order failed to execute
- Timestamps in UTC for all events

Note: If a strategy is active but showing no log entries, it means the signal conditions have not yet been met. Check back after a few minutes.

8. Trade History

The History page shows a comprehensive record of all trades and orders.

8.1 Summary Metrics

Metric	Description
Total Trades	All executed trades (buys + sells)
Realised P&L	Sum of P&L on all SELL trades
Win Rate	% of SELL trades with positive P&L
Open Orders	Count of limit/stop orders currently waiting to fill

8.2 Open Orders Panel

All pending Limit and Stop orders are listed with full details: symbol, side, type, quantity, trigger price, TP, SL, and creation time. Click **Cancel** to remove an order without executing it.

8.3 Executed Trade Log

Column	Description
#	Unique trade ID
Symbol	Trading pair
Side	BUY (green) or SELL (red)
Qty	Number of units traded
Price	Execution price
Value	Qty × Price (gross trade value)
Fee	0.1% taker fee charged for this trade
P&L	Realised profit/loss (SELL trades only; BUY shows —)
Strategy	Name of the automated strategy that triggered the trade, or — for manual trades
Time	Execution timestamp (UTC)

Note: The most recent 100 trades are displayed. All trades are permanently stored in the SQLite database file (~/.currency_trader/currency_trader.db).

9. Settings

9.1 Trading Configuration

Setting	Value	Notes
Fee Rate	0.1%	Fixed taker fee applied to all executed orders
Strategy Interval	30 s	Frequency of automated strategy evaluation
Price Update	1 s	WebSocket push interval for live prices

9.2 Data Sources

Source	Type	Symbols
Binance WebSocket	Live	BTC/USDT ETH/USDT SOL/USDT BNB/USDT ADA/USDT DOGE/USDT XRP/USDT
Forex Simulation	Simulated	EUR/USD GBP/USD USD/JPY AUD/USD USD/CAD USD/CHF NZD/USD

Forex prices use a **mean-reverting random walk** model seeded with realistic exchange rates. The drift pulls prices back toward the seed value, creating natural-looking fluctuations with 0.02% volatility per second.

9.3 Resetting the Paper Portfolio

The **Reset Paper Portfolio** button in the Danger Zone clears all trade history, positions, and orders, returning your balance to the default \$10,000. A confirmation dialog is shown before any data is removed. After resetting, restart the application to apply the changes fully.

■ **Warning:** This action is irreversible. All trade history, P&L; records, and open positions will be permanently deleted from the database.

10. Technical Reference

10.1 Architecture

Layer	Technology	Role
Backend	Python 3.11 + FastAPI	REST API, WebSocket server, trading engine
Database	SQLite +aiosqlite	Persistent storage for trades, portfolio, strategies
Market Data	Binance WebSocket	Live crypto price stream (no API key required)
Frontend	React 18 + TypeScript	Single-page trading terminal UI
Styling	Tailwind CSS	Utility-first dark-theme styling
Charts	TradingView LW Charts	Candlestick, area, and histogram charts
State	Zustand	Lightweight global state management
Build	PyInstaller + Vite	Single-file executable bundling

10.2 API Endpoints

Method	Path	Description
GET	/api/market/prices	All current prices
GET	/api/market/history/{sym}	OHLCV history (up to 200 bars)
GET	/api/market/symbols	List of all supported symbols
GET	/api/portfolio/summary	Portfolio value, positions, P&L
GET	/api/portfolio/performance	Daily P&L, win rate, stats
POST	/api/trades/order	Place a new order
DELETE	/api/trades/order/{id}	Cancel an open order
GET	/api/trades/orders	List orders (filter by status)
GET	/api/trades/history	Executed trade history
GET	/api/strategy/	List all strategies
POST	/api/strategy/	Create a new strategy
PATCH	/api/strategy/{id}	Update strategy (activate/deactivate/params)
DELETE	/api/strategy/{id}	Delete a strategy
GET	/api/strategy/{id}/logs	Strategy execution logs
WS	/ws	WebSocket — live price + portfolio push every 1 s

10.3 Database Schema

Table	Key Columns
portfolio	usd_balance, total_value, total_pnl, total_pnl_pct
positions	symbol, quantity, avg_entry_price, current_price, unrealized_pnl
orders	symbol, side, order_type, quantity, price, stop_price, take_profit, stop_loss, status
trades	symbol, side, quantity, price, value, fee, pnl, strategy, executed_at
strategies	name, strategy_type, symbol, is_active, params(JSON), position_size_pct, stop_loss_pct, take_profit_pct
strategy_logs	strategy_id, message, level, created_at

The database file is located at:

- **Linux / macOS:** ~/.currency_trader/currency_trader.db
- **Windows:** C:\Users\<username>\.currency_trader\currency_trader.db

10.4 Port Configuration

The application binds to **port 8765** by default. If this port is in use, you can modify the port in backend/app/main.py in the run() function and rebuild the executable.

10.5 File Structure

```
currency_trader/ ■■■ backend/ ■ ■■■ app/ ■ ■ ■■■ main.py # FastAPI app + WebSocket ■ ■
■■■ database.py # SQLite async setup ■ ■ ■■■ models.py # ORM models ■ ■ ■■■
market_data.py # Binance + forex prices ■ ■ ■■■ trading_engine.py# Order execution + P&L; ■
■ ■■■ strategies.py # SMA / RSI / MACD / BB ■ ■ ■■■ routers/ # REST endpoints ■ ■■■
requirements.txt ■■■ frontend/ ■ ■■■ src/ ■ ■■■ components/ # React UI components ■ ■■■
store/ # Zustand state ■ ■■■ hooks/ # WebSocket hook ■■■ static/ # Built React app ■■■
build.sh / build.bat # Build scripts ■■■ run_dev.sh # Dev launcher ■■■ currency_trader.spec
# PyInstaller spec
```

11. Troubleshooting

Problem	Likely Cause	Solution
Browser does not open	Port 8765 in use or firewall	Open http://localhost:8765 manually. Check if another process uses port 8765.
Ticker shows '—' or no prices	WebSocket not yet connected	Wait 2–3 seconds for the connection to establish. Check internet access.
Crypto prices not updating	Binance WebSocket blocked	Check firewall / proxy settings. Forex prices work offline.
'Insufficient balance' error	Not enough USD for the trade	Reduce quantity or use the % slider to stay within your balance.
Strategy not placing trades	Signal conditions not met yet	Strategies need sufficient price history (30–50 bars). Wait a few minutes or restart.
Blank white screen	Frontend not built	Run <code>build.sh</code> (Linux) or <code>build.bat</code> (Windows) to compile the frontend.
Windows SmartScreen warning	Binary not code-signed	Click 'More info' → 'Run anyway'. This is expected for self-built binaries.
Database locked error	Multiple instances running	Close all instances of <code>currency_trader</code> and relaunch once.

12. Glossary

Term	Definition
OHLCV	Open, High, Low, Close, Volume — the five values that define a price bar
Candlestick	Visual price bar showing Open, High, Low, Close for a time period
Market Order	An order that executes immediately at the best available price
Limit Order	An order that waits until the market price reaches a specified level
Stop Order	An order that triggers a market execution when price crosses a stop level
Take-Profit (TP)	A price at which an open position is automatically closed for a gain
Stop-Loss (SL)	A price at which an open position is automatically closed to cap a loss
P&L	Profit and Loss — the gain or loss from trading activity
Unrealised P&L	P&L on open positions, calculated at the current market price
Realised P&L	P&L locked in after closing (selling) a position
Position Size	The amount of capital allocated to a single trade
Win Rate	Percentage of closed trades that generated a profit
SMA	Simple Moving Average — arithmetic mean of closing prices over N bars
EMA	Exponential Moving Average — weighted average giving more weight to recent bars
RSI	Relative Strength Index (0–100) — measures momentum and overbought/oversold levels
MACD	Moving Average Convergence Divergence — trend-following momentum indicator
Bollinger Bands	Price channel based on standard deviation around a moving average
Paper Trading	Simulated trading with virtual money — no real financial risk
Taker Fee	Fee paid for orders that consume existing liquidity (market orders)
WebSocket	Full-duplex network protocol enabling real-time server-to-client data push

13. Legal & Disclaimer

■ **Warning:** *Currency Trader Pro is provided for **educational and research purposes only**. It is not financial advice. Past simulated performance does not guarantee future real-world results.*

Currency Trader Pro does not:

- Connect to any real exchange account
- Execute trades with real money
- Hold, transmit, or request any financial credentials
- Provide regulated financial advice

Cryptocurrency and forex markets are highly volatile. Real trading involves substantial risk of loss. Always consult a licensed financial advisor before committing real capital to any trading strategy.

Market data is sourced from Binance's publicly available WebSocket API. The authors are not affiliated with Binance or any other exchange. Forex prices are simulated and do not represent real market conditions.

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